

Practice Quiz: Mixture and Alligations

Competitive Exam Prep

Multiple Choice Questions

1. In what ratio must a grocer mix two varieties of tea costing ₹60 per kg and ₹65 per kg so that by selling the mixture at ₹68.20 per kg he may gain 10%?
A) 3 : 2
B) 3 : 4
C) 3 : 5
D) 4 : 5
2. A container contains 40 litres of milk. From this container, 4 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?
A) 26.34 litres
B) 27.36 litres
C) 28 litres
D) 29.16 litres
3. In what ratio must water be mixed with milk to gain $16\frac{2}{3}\%$ by selling the mixture at cost price?
A) 1 : 6
B) 6 : 1
C) 2 : 3
D) 4 : 3
4. A mixture contains alcohol and water in the ratio 4 : 3. If 5 litres of water is added to the mixture, the ratio becomes 4 : 5. Find the quantity of alcohol in the given mixture.
A) 10 litres
B) 12 litres
C) 15 litres
D) 18 litres
5. 8 kg of tea costing ₹240 per kg is mixed with 9 kg of tea costing ₹250 per kg. The average price per kg of the mixed tea is:
A) ₹245.29
B) ₹246.20
C) ₹244.50
D) ₹248.50

6. A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 25%. The percentage of water in the mixture is:
- 4%
 - 6.25%
 - 20%
 - 25%
7. How many kilograms of sugar costing ₹9 per kg must be mixed with 27 kg of sugar costing ₹7 per kg so that there may be a gain of 10% by selling the mixture at ₹9.24 per kg?
- 36 kg
 - 42 kg
 - 54 kg
 - 63 kg
8. In two types of brass, the ratios of Copper to Zinc are 8 : 3 and 15 : 7 respectively. If the two types are melted and mixed in the ratio 5 : 2, a new type of brass is formed. The ratio of Copper to Zinc in this new brass is:
- 2 : 1
 - 3 : 1
 - 5 : 2
 - 11 : 5
9. A jar full of whisky contains 40% alcohol. A part of this whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%. The quantity of whisky replaced is:
- $\frac{1}{3}$
 - $\frac{2}{3}$
 - $\frac{2}{5}$
 - $\frac{3}{5}$
10. A vessel is filled with liquid, 3 parts of which are water and 5 parts syrup. How much of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup?
- $\frac{1}{3}$
 - $\frac{1}{4}$
 - $\frac{1}{5}$
 - $\frac{1}{7}$
11. Two vessels A and B contain milk and water mixed in the ratio 8 : 5 and 5 : 2 respectively. The ratio in which these two mixtures be mixed to get a new mixture containing $69\frac{3}{13}\%$ milk is:
- 2 : 7
 - 3 : 5
 - 5 : 2
 - 5 : 7

12. Milk and water in two vessels A and B are in the ratio 4 : 3 and 2 : 3 respectively. In what ratio the liquids in both the vessels are mixed to obtain a new mixture in vessel C containing half milk and half water?
- A) 7 : 5
B) 5 : 7
C) 3 : 11
D) 1 : 1
13. A merchant has 1000 kg of sugar, part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. The quantity sold at 18% profit is:
- A) 400 kg
B) 560 kg
C) 600 kg
D) 640 kg
14. A can contains a mixture of two liquids A and B in the ratio 7 : 5. When 9 litres of mixture are drawn off and the can is filled with B, the ratio of A and B becomes 7 : 9. How many litres of liquid A was contained by the can initially?
- A) 10 litres
B) 20 litres
C) 21 litres
D) 25 litres
15. One type of liquid contains 20% water and the second type contains 35% water. A glass is filled with 10 parts of first liquid and 4 parts of second liquid. The percentage of water in the new mixture in the glass is:
- A) 20%
B) $24\frac{2}{7}\%$
C) 37%
D) 40%
16. Gold is 19 times as heavy as water and copper is 9 times as heavy as water. In what ratio should these be mixed to get an alloy 15 times as heavy as water?
- A) 1 : 1
B) 2 : 3
C) 3 : 2
D) 4 : 3
17. In what ratio must a grocer mix two varieties of pulses costing ₹15 and ₹20 per kg respectively so as to get a mixture worth ₹16.50 per kg?
- A) 3 : 7
B) 7 : 3
C) 5 : 2
D) 2 : 5

18. A sum of ₹36.90 is made up of 90 coins that are either 20 paise or 50 paise coins. How many 20 paise coins are there?
- A) 27
 - B) 43
 - C) 63
 - D) 72
19. A zookeeper counted the heads of the animals in a cage and found 80. When he counted the legs, he found 260. If the cage contains only rabbits and pigeons, how many rabbits are there?
- A) 30
 - B) 50
 - C) 60
 - D) 70
20. A mixture of 150 litres of wine and water contains 20% water. How much more water should be added so that water becomes 25% of the new mixture?
- A) 5 litres
 - B) 10 litres
 - C) 15 litres
 - D) 20 litres
21. In an alloy, zinc and copper are in the ratio 1 : 2. In the second alloy, the same elements are in the ratio 2 : 3. If these two alloys are mixed to form a new alloy in which two elements are in the ratio 5 : 8, the ratio of these two alloys in the new alloy is:
- A) 3 : 10
 - B) 3 : 7
 - C) 10 : 3
 - D) 7 : 3
22. A milkman has two cans. The first can contains 25% water and the rest milk. The second can contains 50% water and the rest milk. How much milk should he mix from each container so as to get 12 litres of a mixture such that the ratio of water to milk is 3 : 5?
- A) 4 litres, 8 litres
 - B) 6 litres, 6 litres
 - C) 5 litres, 7 litres
 - D) 3 litres, 9 litres
23. 400 grams of sugar solution has 40% sugar in it. How much sugar should be added to make it 50% in the solution?
- A) 40g
 - B) 60g
 - C) 80g
 - D) 100g

- 24.** A container has 80 litres of milk. From this container 8 litres of milk was taken out and replaced by water. The process was further repeated twice. How much milk is now left in the container?
- A) 46.52 litres
B) 58.32 litres
C) 64.80 litres
D) 60 litres
- 25.** A vessel contains 20 litres of a mixture of milk and water in the ratio 3 : 2. 10 litres of the mixture are removed and replaced with an equal quantity of pure milk. If the process is repeated once more, the ratio of milk and water in the final mixture is:
- A) 1 : 4
B) 4 : 1
C) 9 : 1
D) 19 : 1
- 26.** ₹6.40 per kg of rice is mixed with ₹8 per kg rice. The ratio of the quantities was 1 : 3. Find the cost of the mixture per kg.
- A) ₹7.60
B) ₹7.50
C) ₹7.40
D) ₹7.20
- 27.** In what ratio must a merchant mix two kinds of rice (one at ₹1.15 per kg and another at ₹1.24 per kg) so that the mixture may be worth ₹1.21 per kg?
- A) 1 : 2
B) 3 : 4
C) 4 : 5
D) 2 : 1
- 28.** How much water must be added to 60 litres of milk at 1.5 litres for ₹20 so as to have a mixture worth ₹ $10\frac{2}{3}$ per litre?
- A) 10 litres
B) 15 litres
C) 20 litres
D) 25 litres
- 29.** In a mixture of 60 litres, the ratio of milk and water is 2 : 1. If this ratio is to be 1 : 2, then the quantity of water to be further added is:
- A) 20 litres
B) 30 litres
C) 40 litres
D) 60 litres

30. A vessel contains milk and water in the ratio 3 : 2. Another vessel contains milk and water in the ratio 2 : 3. In what ratio should the contents of the two vessels be mixed such that the new mixture contains milk and water in the ratio 1 : 1?

- A) 1 : 1
- B) 2 : 3
- C) 3 : 2
- D) 4 : 5